

Math Sec 1

1. Let $y = y(x)$ be the solution of the differential equation $x \frac{dy}{dx} - 2y = 2 + 3 \sin x$, $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$, $y(0) = -\frac{7}{4}$. Then $y(\frac{\pi}{6})$ is equal to :
 (A) $-3\sqrt{2} - 7$ (B) $-3\sqrt{3} - 7$
 (C) $-\frac{5}{4}$ (D) $-\frac{5}{2}$
2. Let $A = \{2, 3, 5, 7, 9\}$. Let R be the relation on A defined by xRy if and only if $2x \leq 3y$. Let l be the number of elements in R , and m be the minimum number of elements required to be added in R to make it a symmetric relation. Then $l + m$ is equal to:
 (A) 27 (B) 21 (C) 25 (D) 23
3. Let the line L_1 be parallel to the vector $-3\hat{i} + 2\hat{j} + 4\hat{k}$ and pass through the point $(2, 6, 7)$, and the line L_2 be parallel to the vector $2\hat{i} + \hat{j} + 3\hat{k}$ and pass through the point $(4, 3, 5)$. If the line L_3 is parallel to the vector $-3\hat{i} + 5\hat{j} + 16\hat{k}$ and intersects the lines L_1 and L_2 at the points C and D , respectively, then $|\overrightarrow{CD}|^2$ is equal to:
 (A) 312 (B) 290 (C) 171 (D) 89
4. Let one end of a focal chord of the parabola $y^2 = 16x$ be $(16, 16)$. If $P(\alpha, \beta)$ divides this focal chord internally in the ratio $5 : 2$, then the minimum value of $\alpha + \beta$ is equal to :
 (A) 7 (B) 5 (C) 16 (D) 22
5. Let α and β be the roots of the equation $x^2 + 2ax + (3a + 10) = 0$ such that $\alpha < 1 < \beta$. Then the set of all possible values of a is :
 (A) $(-\infty, \frac{-11}{5}) \cup (5, \infty)$
 (B) $(-\infty, -2) \cup (5, \infty)$
 (C) $(-\infty, -3)$
 (D) $(-\infty, \frac{-11}{5})$
6. For the matrices $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} -29 & 49 \\ -13 & 18 \end{bmatrix}$, if $(A^{15} + B) \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, then among the following which one is true?
 (A) $x = 5, y = 7$ (B) $x = 16, y = 3$
 (C) $x = 18, y = 11$ (D) $x = 11, y = 2$
7. Let $y^2 = 12x$ be the parabola with its vertex at O . Let P be a point on the parabola and A be a point on the x -axis such that $\angle OPA = 90^\circ$. Then the locus of the centroid of such triangles OPA is :
 (A) $y^2 - 4x + 8 = 0$ (B) $y^2 - 9x + 6 = 0$
 (C) $y^2 - 6x + 4 = 0$ (D) $y^2 - 2x + 8 = 0$
8. Let $f : \mathbf{R} \rightarrow \mathbf{R}$ be a twice differentiable function such that $f''(x) > 0$ for all $x \in \mathbf{R}$ and $f'(a - 1) = 0$, where a is a real number. Let $g(x) = f(\tan^2 x - 2 \tan x + a)$, $0 < x < \frac{\pi}{2}$.
 Consider the following two statements:
 (I) g is increasing in $(0, \frac{\pi}{4})$
 (II) g is decreasing in $(\frac{\pi}{4}, \frac{\pi}{2})$
 Then,
 (A) Neither (I) nor (II) is True (B) Only (II) is True
 (C) Both (I) and (II) are True (D) Only (I) is True
9. If the system of equations
 $3x + y + 4z = 3$
 $2x + \alpha y - z = -3$
 $x + 2y + z = 4$
 has no solution, then the value of α is equal to:
 (A) 13 (B) 19 (C) 4 (D) 23
10. Let the line L pass through the point $(-3, 5, 2)$ and make equal angles with the positive coordinate axes. If the distance of L from the point $(-2, r, 1)$ is $\sqrt{\frac{14}{3}}$, then the sum of all possible values of r is:
 (A) 6 (B) 10 (C) 12 (D) 16
11. If the line $\alpha x + 4y = \sqrt{7}$, where $\alpha \in \mathbf{R}$, touches the ellipse $3x^2 + 4y^2 = 1$ at the point P in the first quadrant, then one of the focal distances of P is:
 (A) $\frac{1}{\sqrt{3}} + \frac{1}{2\sqrt{7}}$
 (B) $\frac{1}{\sqrt{3}} - \frac{1}{2\sqrt{11}}$
 (C) $\frac{1}{\sqrt{3}} + \frac{1}{2\sqrt{5}}$
 (D) $\frac{1}{\sqrt{3}} - \frac{1}{2\sqrt{5}}$
12. Let $A = \{x : |x^2 - 10| \leq 6\}$ and $B = \{x : |x - 2| > 1\}$. Then
 (A) $A \cap B = [-4, -2] \cup [3, 4]$
 (B) $A \cup B = (-\infty, 1] \cup (2, \infty)$
 (C) $B - A = (-\infty, -4) \cup (-2, 1) \cup (4, \infty)$
 (D) $A - B = [2, 3]$
13. For a triangle ABC , let $\vec{p} = \overrightarrow{BC}$, $\vec{q} = \overrightarrow{CA}$ and $\vec{r} = \overrightarrow{BA}$. If $|\vec{p}| = 2\sqrt{3}$, $|\vec{q}| = 2$ and $\cos \theta = \frac{1}{\sqrt{3}}$, where θ is the angle between $\vec{p}$ and $\vec{q}$, then $|\vec{p} \times (\vec{q} - 3\vec{r})|^2 + 3|\vec{r}|^2$ is equal to:
 (A) 410 (B) 340 (C) 220 (D) 200

14. Let $a_1, \frac{a_2}{2}, \frac{a_3}{2^2}, \dots, \frac{a_{10}}{2^9}$ be a G.P. of common ratio $\frac{1}{\sqrt{2}}$. If $a_1 + a_2 + \dots + a_{10} = 62$, then a_1 is equal to :
- (A) $2(2 - \sqrt{2})$ (B) $\sqrt{2} - 1$
 (C) $2 - \sqrt{2}$ (D) $2(\sqrt{2} - 1)$
15. The largest $n \in \mathbf{N}$, for which 7^n divides $101!$, is :
- (A) 19 (B) 15 (C) 16 (D) 18
16. Let z be the complex number satisfying $|z - 5| \leq 3$ and having maximum positive principal argument. Then $34 \left| \frac{5z-12}{5iz+16} \right|^2$ is equal to :
- (A) 12 (B) 16 (C) 26 (D) 20
17. A random variable X takes values 0, 1, 2, 3 with probabilities $\frac{2a+1}{30}, \frac{8a-1}{30}, \frac{4a+1}{30}, b$ respectively, where $a, b \in \mathbf{R}$. Let μ and σ respectively be the mean and standard deviation of X such that $\sigma^2 + \mu^2 = 2$. Then $\frac{a}{b}$ is equal to :

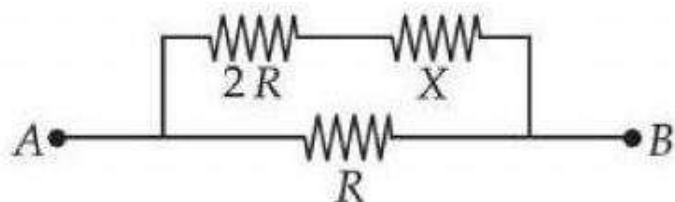
- (A) 30 (B) 60 (C) 12 (D) 3
18. If the area of the region $\{(x, y) : 1 - 2x \leq y \leq 4 - x^2, x \geq 0, y \geq 0\}$ is $\frac{a}{\beta}$, $a, \beta \in \mathbf{N}$, $\gcd(a, \beta) = 1$, then the value of $(\alpha + \beta)$ is:
- (A) 67 (B) 73 (C) 91 (D) 85
19. Let $f(x) = x^3 + x^2 f'(1) + 2x f''(2) + f'''(3)$, $x \in \mathbf{R}$. Then the value of $f'(5)$ is:
- (A) $\frac{117}{5}$ (B) $\frac{657}{5}$
 (C) $\frac{2}{5}$ (D) $\frac{62}{5}$
20. The positive integer n , for which the solutions of the equation $x(x+2) + (x+2)(x+4) + \dots + (x+2n-2)(x+2n) = \frac{8n}{3}$ are two consecutive even integers, is :
- (A) 3 (B) 9 (C) 6 (D) 12

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21. If $\left(\frac{1}{{}^{15}C_0} + \frac{1}{{}^{15}C_1}\right) \left(\frac{1}{{}^{15}C_1} + \frac{1}{{}^{15}C_2}\right) \dots \left(\frac{1}{{}^{15}C_{12}} + \frac{1}{{}^{15}C_{13}}\right) = \frac{a^{13}}{{}^{14}C_0 {}^{14}C_1 \dots {}^{14}C_{12}}$, then $30a$ is equal to
22. If $\int_0^1 4 \cot^{-1}(1 - 2x + 4x^2) dx = a \tan^{-1}(2) - b \log_e(5)$, where $a, b \in \mathbf{N}$, then $(2a + b)$ is equal to ____.
23. Let $[\cdot]$ denote the greatest integer function and $f(x) = \lim_{n \rightarrow \infty} \frac{1}{n^3} \sum_{k=1}^n \left[\frac{k^2}{3^x}\right]$. Then $12 \sum_{j=1}^{\infty} f(j)$ is equal to ____.
24. Let the maximum value of $(\sin^{-1} x)^2 + (\cos^{-1} x)^2$ for $x \in \left[-\frac{\sqrt{3}}{2}, \frac{1}{\sqrt{2}}\right]$ be $\frac{m}{n}\pi^2$, where $\gcd(m, n) = 1$. Then $m + n$ is equal to ____.
25. If P is a point on the circle $x^2 + y^2 = 4$, Q is a point on the straight line $5x + y + 2 = 0$ and $x - y + 1 = 0$ is the perpendicular bisector of PQ , then 13 times the sum of abscissa of all such points P is

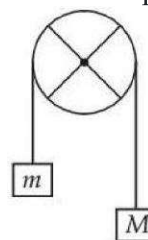
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26. Two known resistances of $R\Omega$ and $2R\Omega$ and one unknown resistance $X\Omega$ are connected in a circuit as shown in the figure. If the equivalent resistance between points A and B in the circuit is $X\Omega$, then the value of X is _____ Ω .



- (A) $(\sqrt{3} + 1)R$ (B) $(\sqrt{3} - 1)R$
 (C) $2(\sqrt{3} - 1)R$ (D) R

27. The pulley shown in figure is made using a thin rim and two rods of length equal to diameter of the rim. The rim and each rod have a mass of M . Two blocks of mass of M and m are attached to two ends of a light string passing over the pulley, which is hinged to rotate freely in vertical plane about its center. The magnitudes of the acceleration experienced by the blocks is _____ (assume no slipping of string on pulley).



- (A) $\frac{(M-m)g}{2M+m}$
 (B) $\frac{(M-m)g}{M+m}$
 (C) $\frac{(M-m)g}{\left[\left(\frac{8}{3}\right)M+m\right]}$
 (D) $\frac{(M-m)g}{\left[\left(\frac{13}{6}\right)M+m\right]}$